LAB EXPERIMENT RECORD

Set 17: Employee Performance Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
Date	13 October 2026
Day	Tuesday

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

1. Line Chart

Aim

To create a line chart to visualize the performance trend of employees over their years of service and label the axes and title the chart.

R Code

```
years <- c(5, 3, 7)

performance <- c(85, 92, 78)

plot(years, performance,

    type = "o",

    pch = 19,
```

col = "blue",

main = "Employee Performance Trend",

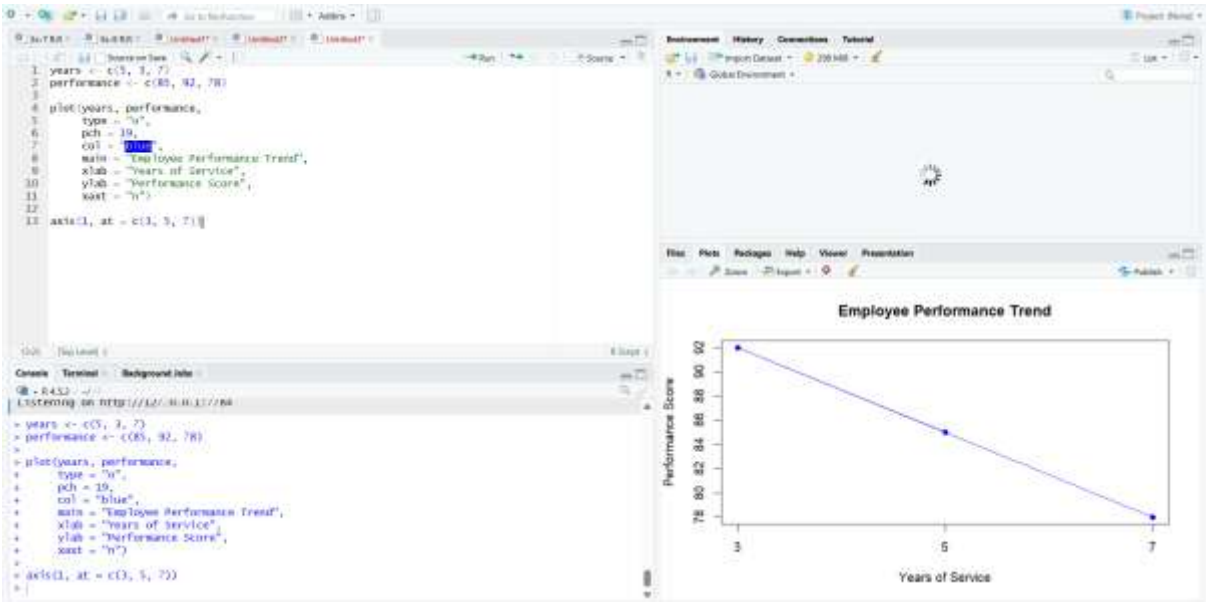
xlab = "Years of Service",

ylab = "Performance Score",

xaxt = "n")

axis(1, at = c(3, 5, 7))

Output



A line chart showing employee performance scores according to their years of service.

Observation

Employee 2 has the highest performance score of 92 with 3 years of service. Employee 3 has the lowest performance score of 78 with 7 years of service.

2. Bar Chart

Aim

To create a bar chart showing the distribution of employees across different departments.

R Code

department <- c("Sales", "HR", "Marketing")

employee_count <- c(1, 1, 1)

barplot(employee_count,

names.arg = department,

main = "Distribution of Employees by Department",

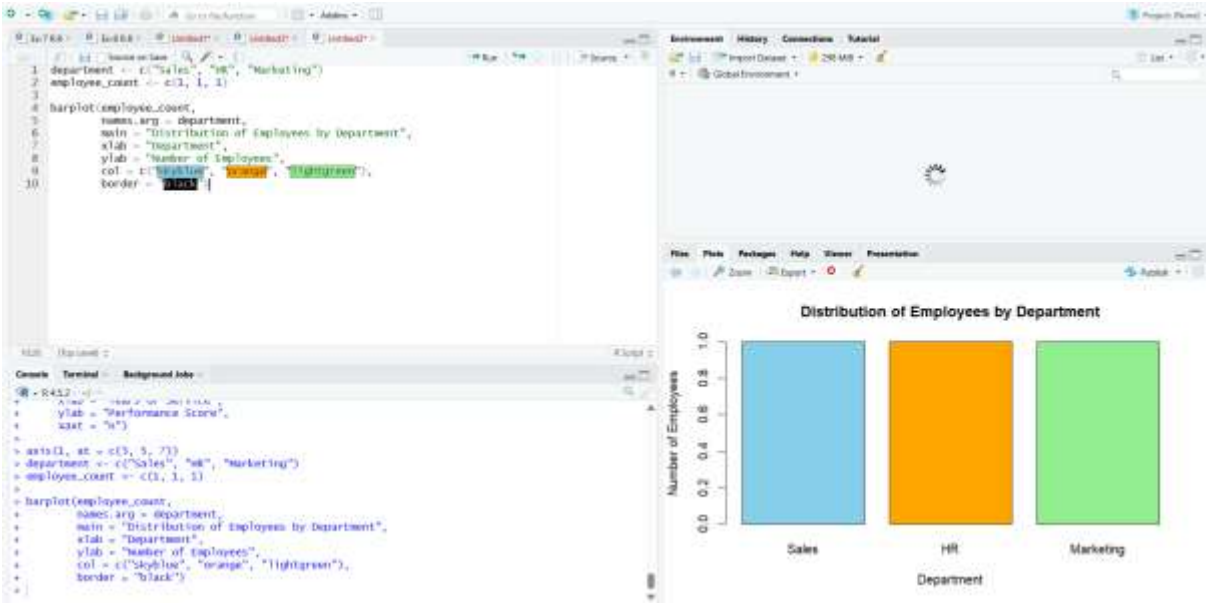
xlab = "Department",

ylab = "Number of Employees",

col = c("skyblue", "orange", "lightgreen"),

border = "black")

Output



A bar chart showing the number of employees in the Sales, HR, and Marketing departments.

Observation

Each department has one employee. Therefore, all three departments have an equal employee count.

3. Employee Performance Table

Aim

To create a table displaying the performance data of each employee, including employee ID, department, years of service, and performance score.

R Code

employee_data <- data.frame(

Employee_ID = c(1, 2, 3),

Department = c("Sales", "HR", "Marketing"),

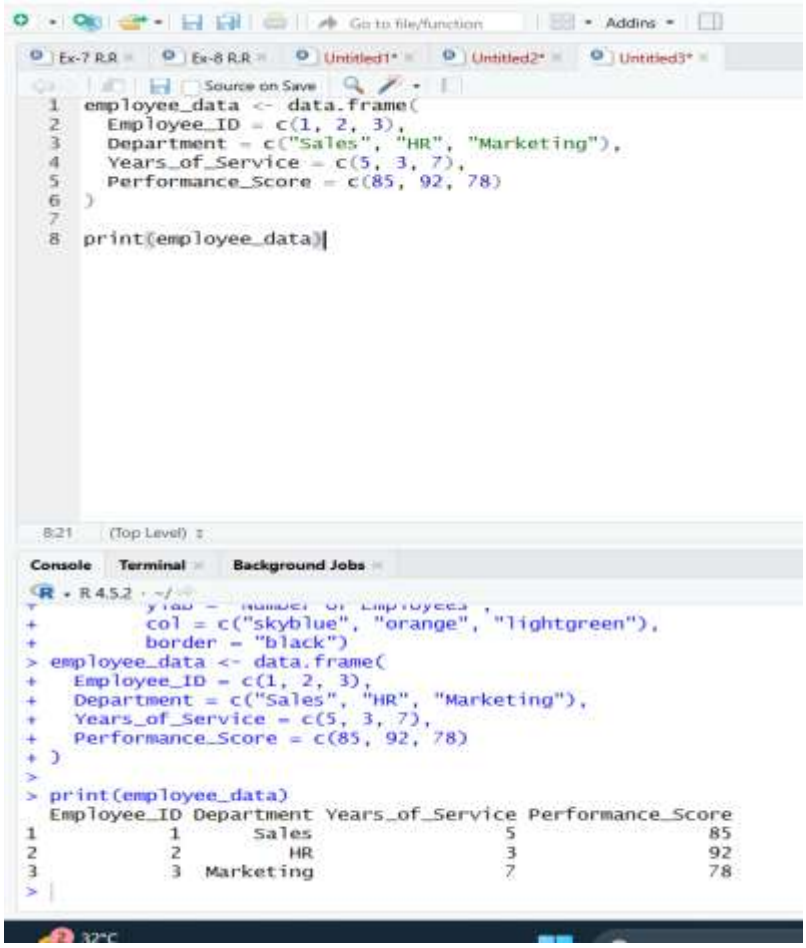
Years_of_Service = c(5, 3, 7),

Performance_Score = c(85, 92, 78)

)

print(employee_data)

Output



Observation

The table clearly displays the performance information of all three employees. Employee 2 has the highest performance score, while Employee 3 has the lowest performance score.

4. Tableau Dashboard

Aim

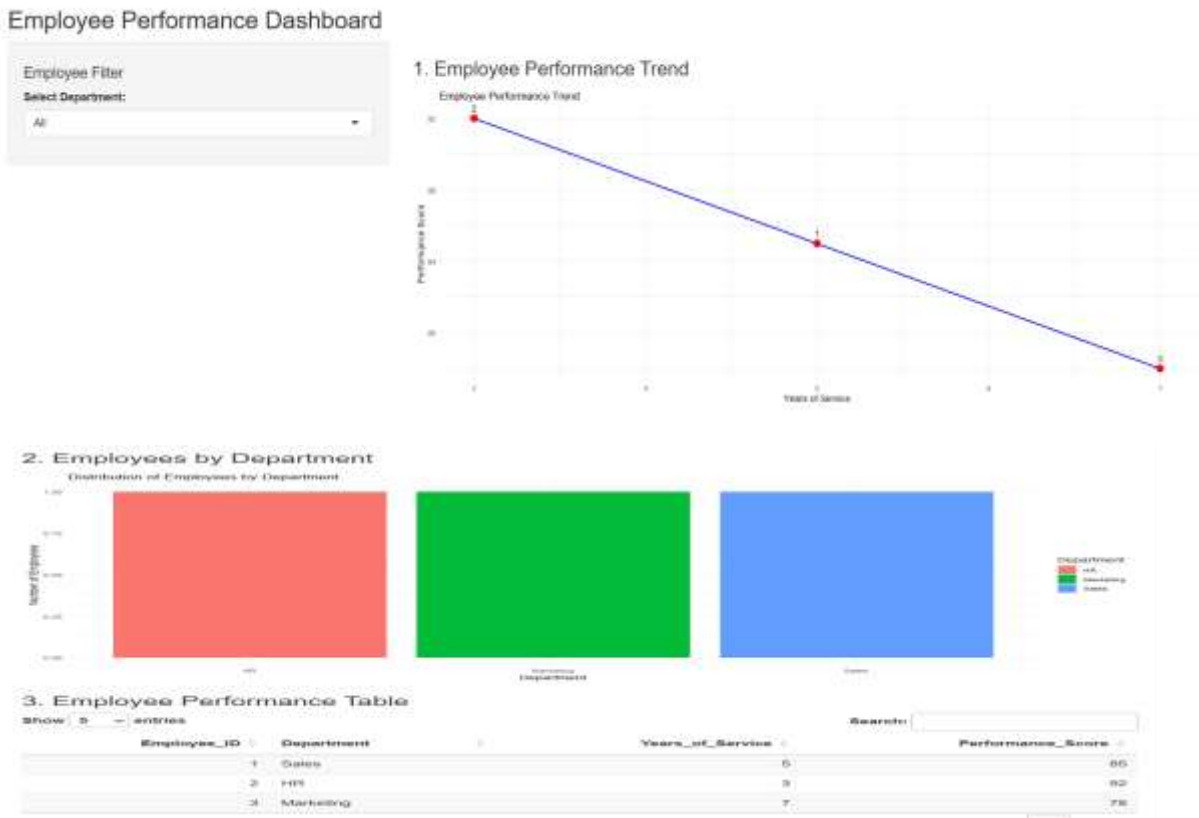
To develop an interactive dashboard combining the line chart, bar chart, and employee performance table for exploring employee performance data.

Tableau Procedure

1. Open Tableau.

2. Import the Employee Performance dataset.
3. Create a line chart using Years of Service and Performance Score.
4. Add suitable chart titles and axis labels.
5. Create a bar chart using Department and the count of Employee ID.
6. Create a table containing Employee ID, Department, Years of Service, and Performance Score.
7. Select Dashboard → New Dashboard.
8. Drag the line chart into the dashboard.
9. Drag the department bar chart into the dashboard.
10. Drag the employee performance table into the dashboard.
11. Arrange all charts and the table neatly.
12. Add suitable titles and labels.
13. Add filters if required for interactive exploration.
14. Save the dashboard.

Output



An interactive dashboard displaying the employee performance trend, department-wise employee distribution, and employee performance details.

Result

The employee performance dataset was successfully analyzed using a line chart, bar chart, and table in R. A Tableau dashboard was designed to provide an interactive view of employee performance and department-wise distribution.